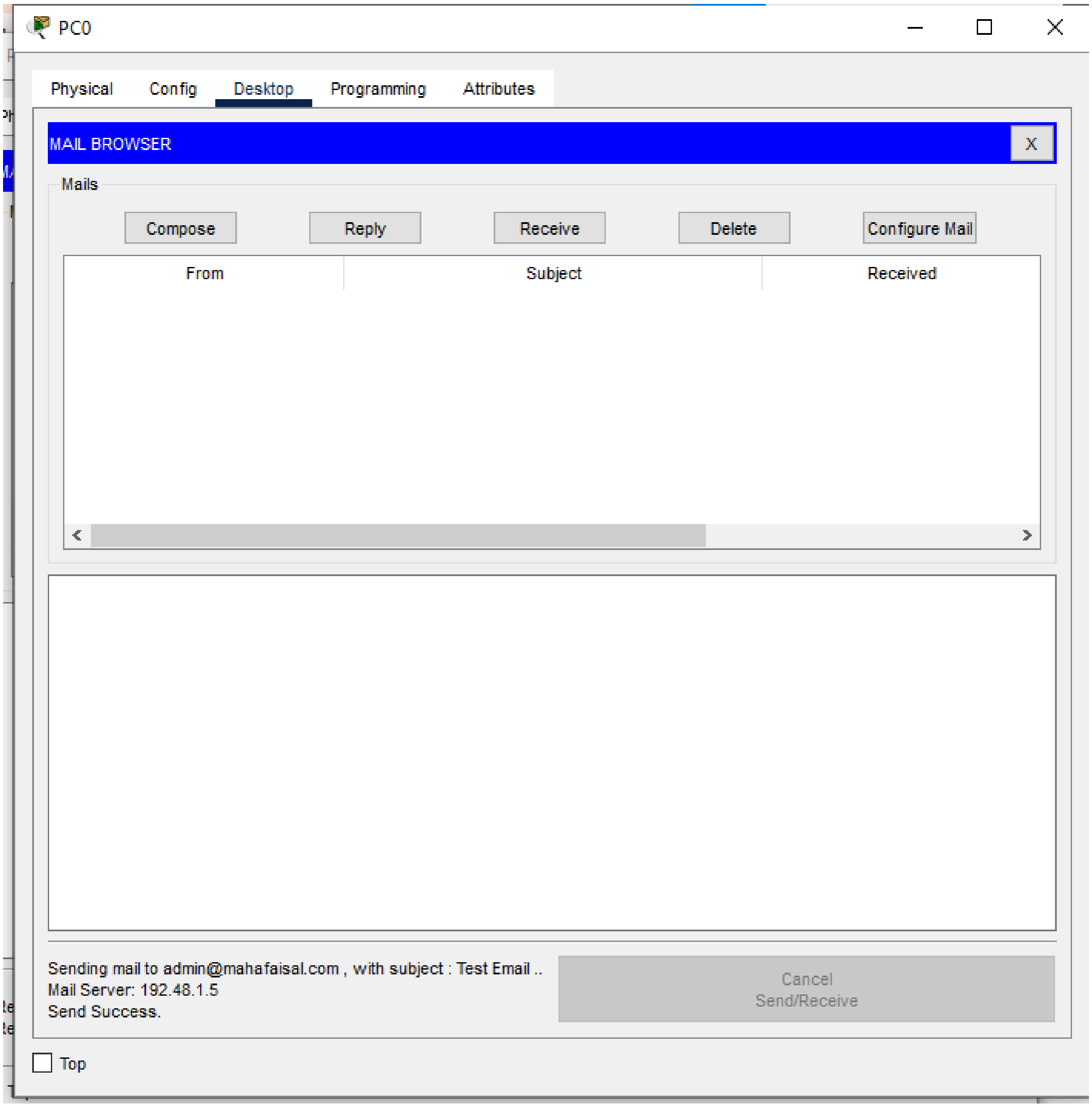
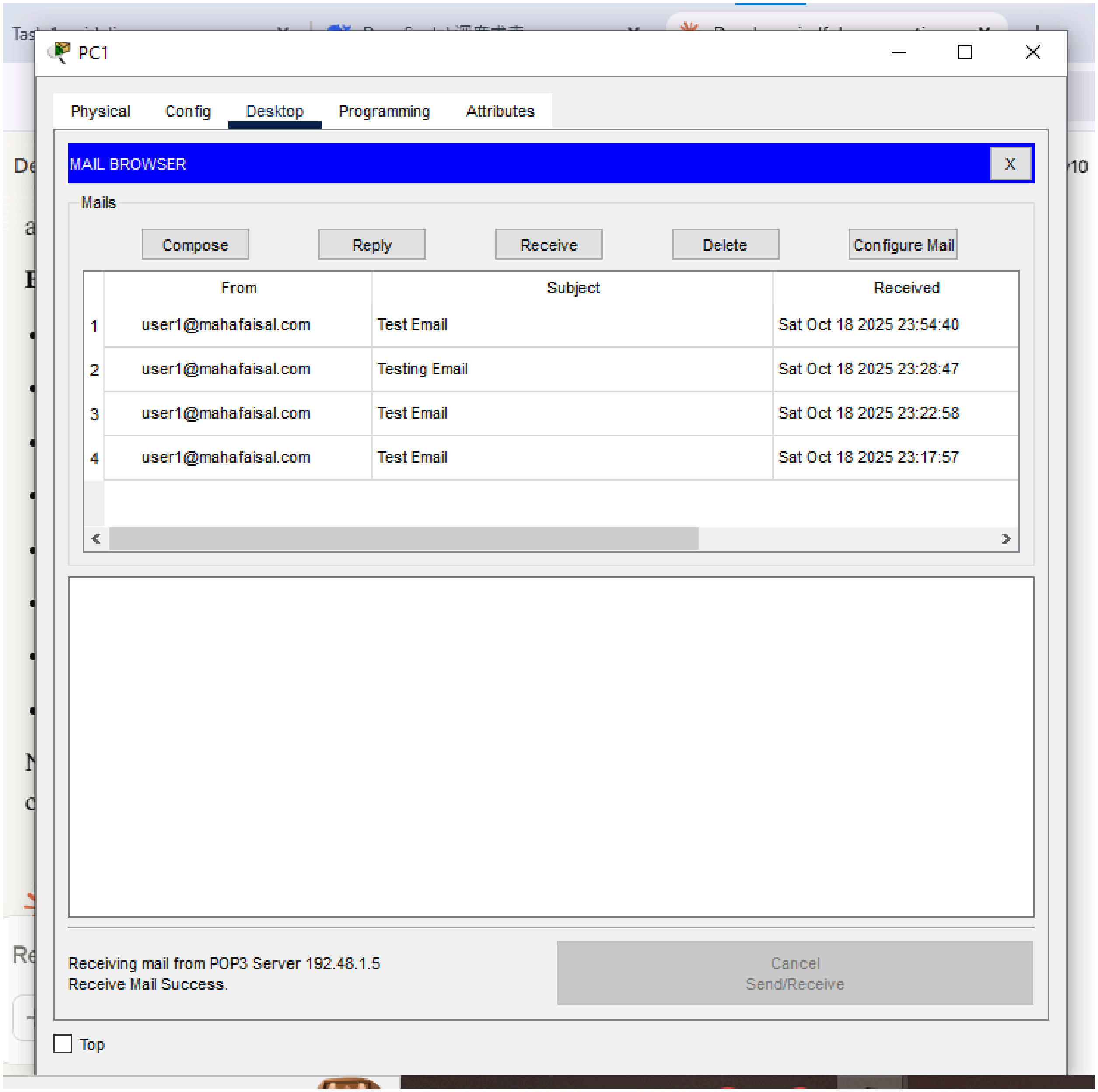
Computer Networks Project 1

Name: Maha Faisal
Roll-Number: Bscs24048

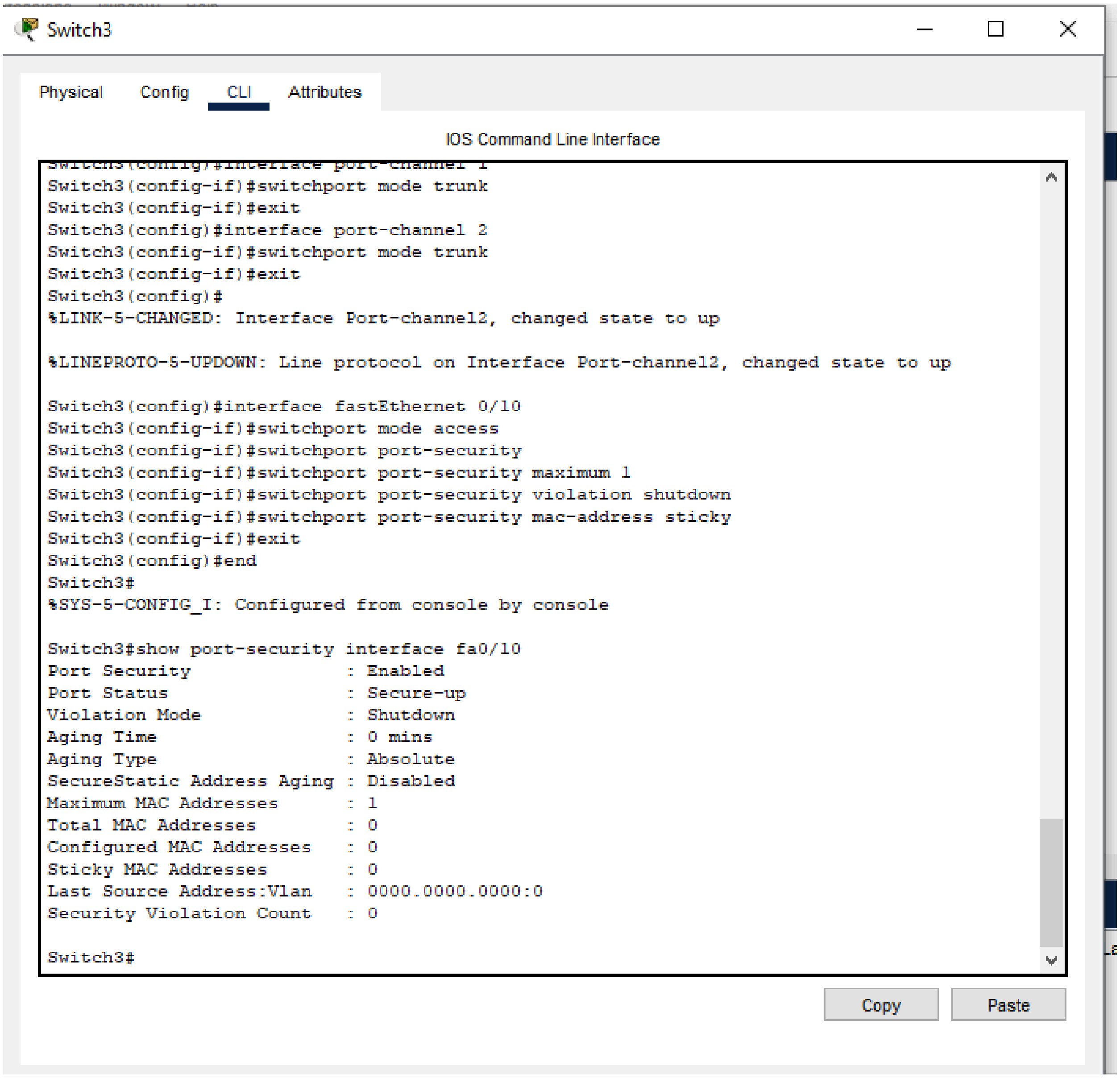
Block 1:




PC1
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
1 user1@mahafaisal.com Test Email Sat Oct 18 2025 23:54:40
2 user1@mahafaisal.com Testing Email Sat Oct 18 2025 23:28:47
3 user1@mahafaisal.com Test Email Sat Oct 18 2025 23:22:58
4 user1@mahafaisal.com Test Email Sat Oct 18 2025 23:17:57
Receiving mail from POP3 Server 192.48.1.5
Receive Mail Success.
Cancel
Send/Receive
Top

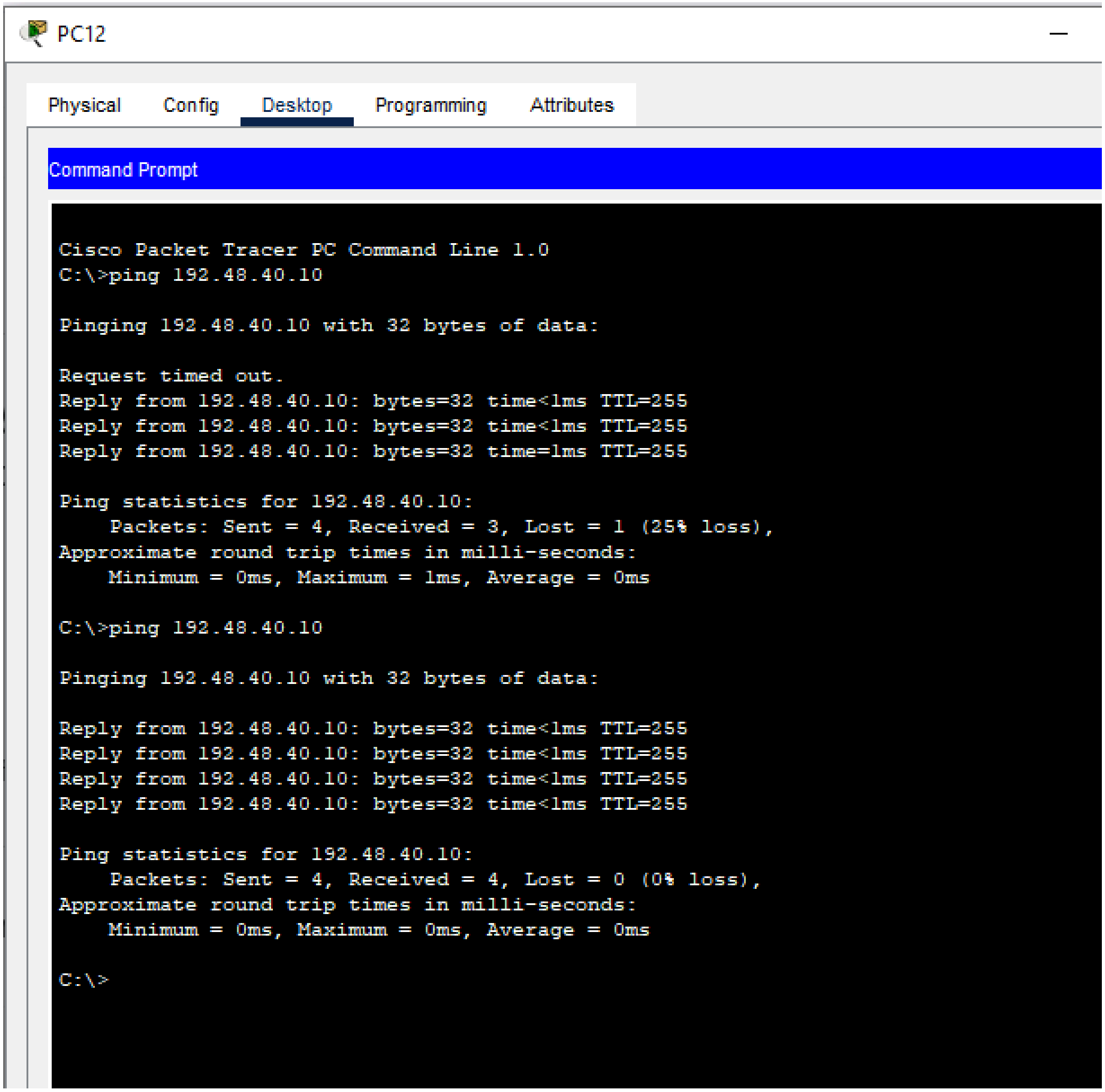

PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://mahafaisal.com
Go
Stop
Name: Maha Faisal
Roll No: Bscs24048
Top

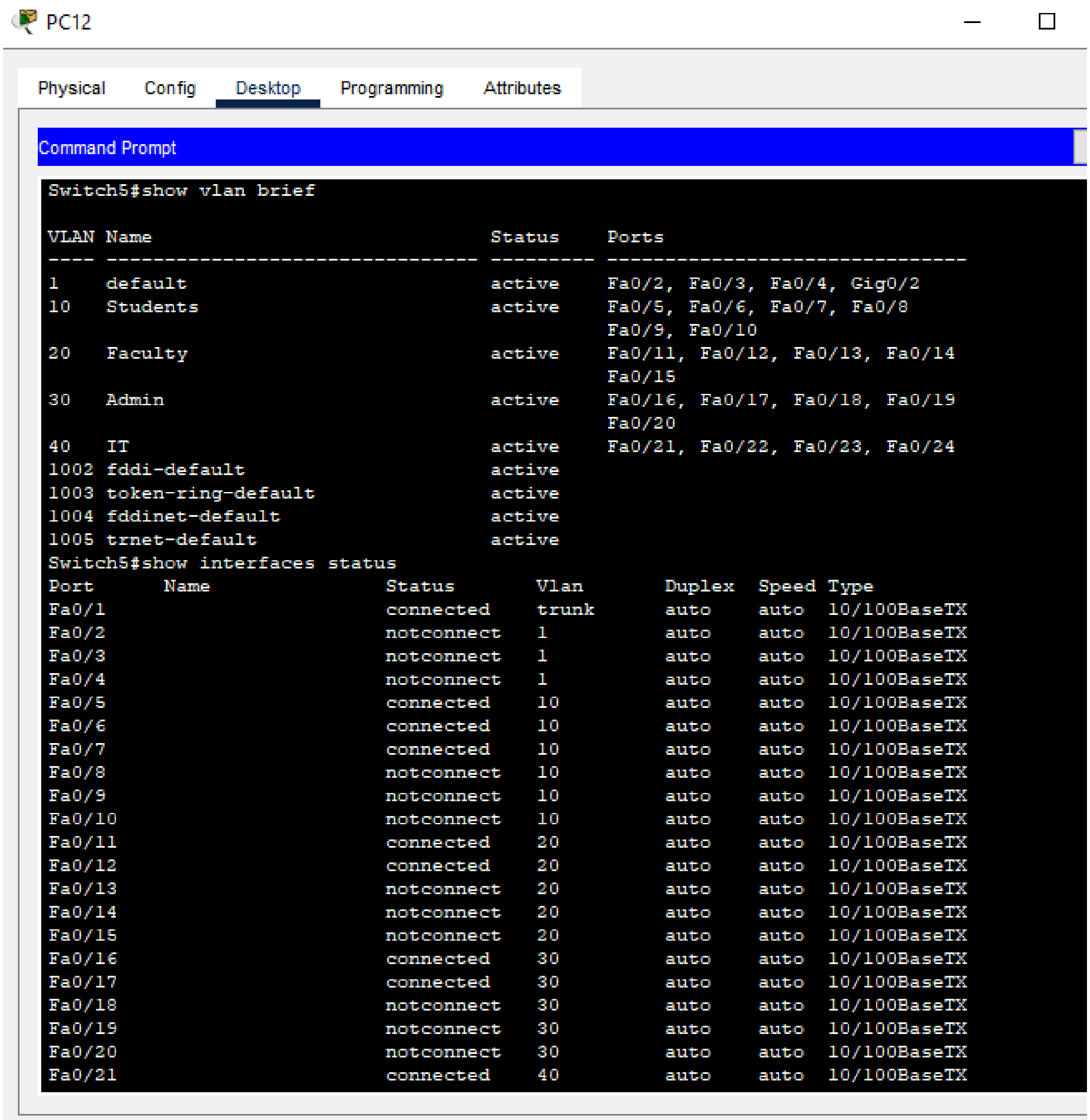
Block 2:



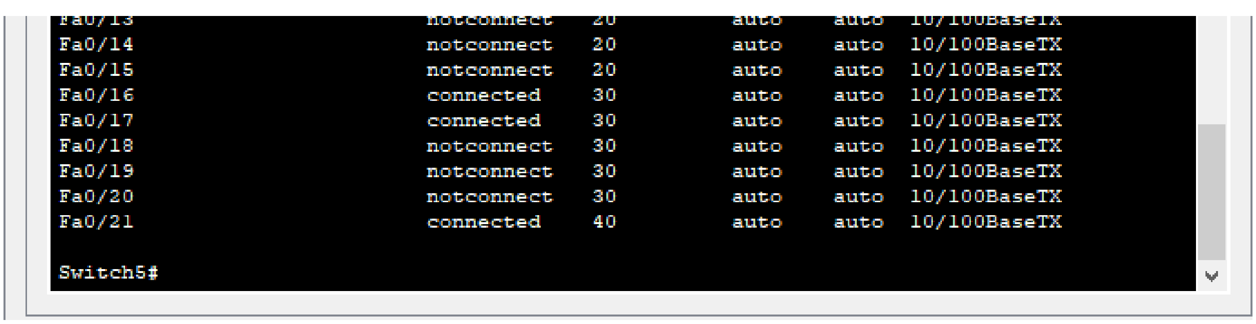

```
Switch3(config)#interface port-channel 1
Switch3(config-if)#switchport mode trunk
Switch3(config-if)#exit
Switch3(config)#interface port-channel 2
Switch3(config-if)#switchport mode trunk
Switch3(config-if)#exit
Switch3(config)#
%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

Switch3(config)#interface fastEthernet 0/10
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport port-security
Switch3(config-if)#switchport port-security maximum 1
Switch3(config-if)#switchport port-security violation shutdown
Switch3(config-if)#switchport port-security mac-address sticky
Switch3(config-if)#exit
Switch3(config)#end
Switch3#
%SYS-5-CONFIG_I: Configured from console by console

Switch3#show port-security interface fa0/10
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0

Switch3#
```

Block 3:


PC12
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.48.40.10
Pinging 192.48.40.10 with 32 bytes of data:
Request timed out.
Reply from 192.48.40.10: bytes=32 time<1ms TTL=255
Reply from 192.48.40.10: bytes=32 time<1ms TTL=255
Reply from 192.48.40.10: bytes=32 time=1ms TTL=255
Ping statistics for 192.48.40.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.48.40.10
Pinging 192.48.40.10 with 32 bytes of data:
Reply from 192.48.40.10: bytes=32 time<1ms TTL=255
Reply from 192.48.40.10: bytes=32 time<1ms TTL=255
Reply from 192.48.40.10: bytes=32 time<1ms TTL=255
Reply from 192.48.40.10: bytes=32 time<1ms TTL=255
Ping statistics for 192.48.40.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

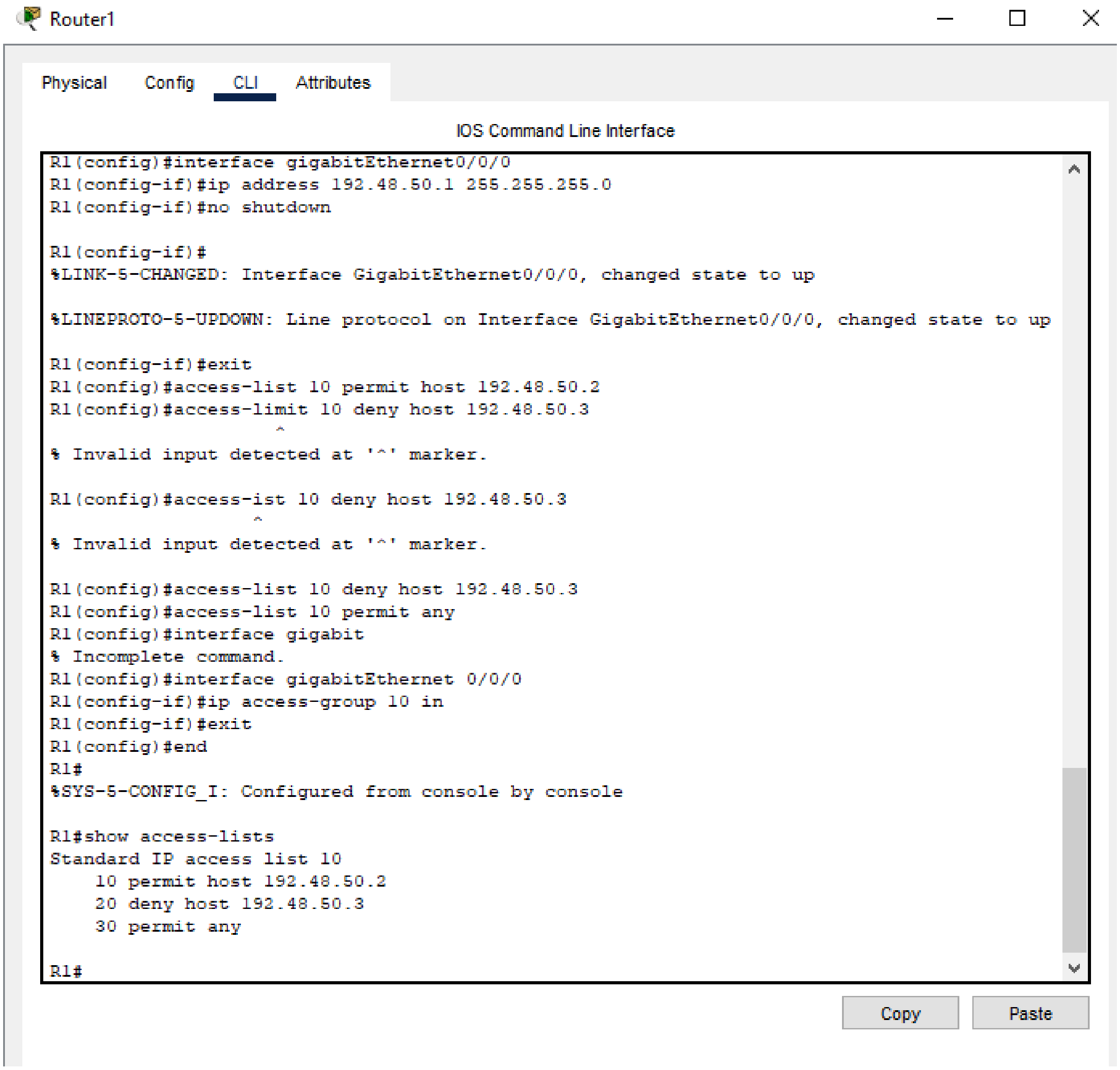



```
Switch5#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Gig0/2
10   Students                         active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
20   Faculty                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15
30   Admin                            active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20
40   IT                               active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch5#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    trunk      auto    auto  10/100BaseTX
Fa0/2                        notconnect   1          auto    auto  10/100BaseTX
Fa0/3                        notconnect   1          auto    auto  10/100BaseTX
Fa0/4                        notconnect   1          auto    auto  10/100BaseTX
Fa0/5                        connected    10         auto    auto  10/100BaseTX
Fa0/6                        connected    10         auto    auto  10/100BaseTX
Fa0/7                        connected    10         auto    auto  10/100BaseTX
Fa0/8                        notconnect   10         auto    auto  10/100BaseTX
Fa0/9                        notconnect   10         auto    auto  10/100BaseTX
Fa0/10                       notconnect   10         auto    auto  10/100BaseTX
Fa0/11                       connected    20         auto    auto  10/100BaseTX
Fa0/12                       connected    20         auto    auto  10/100BaseTX
Fa0/13                       notconnect   20         auto    auto  10/100BaseTX
Fa0/14                       notconnect   20         auto    auto  10/100BaseTX
Fa0/15                       notconnect   20         auto    auto  10/100BaseTX
Fa0/16                       connected    30         auto    auto  10/100BaseTX
Fa0/17                       connected    30         auto    auto  10/100BaseTX
Fa0/18                       notconnect   30         auto    auto  10/100BaseTX
Fa0/19                       notconnect   30         auto    auto  10/100BaseTX
Fa0/20                       notconnect   30         auto    auto  10/100BaseTX
Fa0/21                       connected    40         auto    auto  10/100BaseTX
```

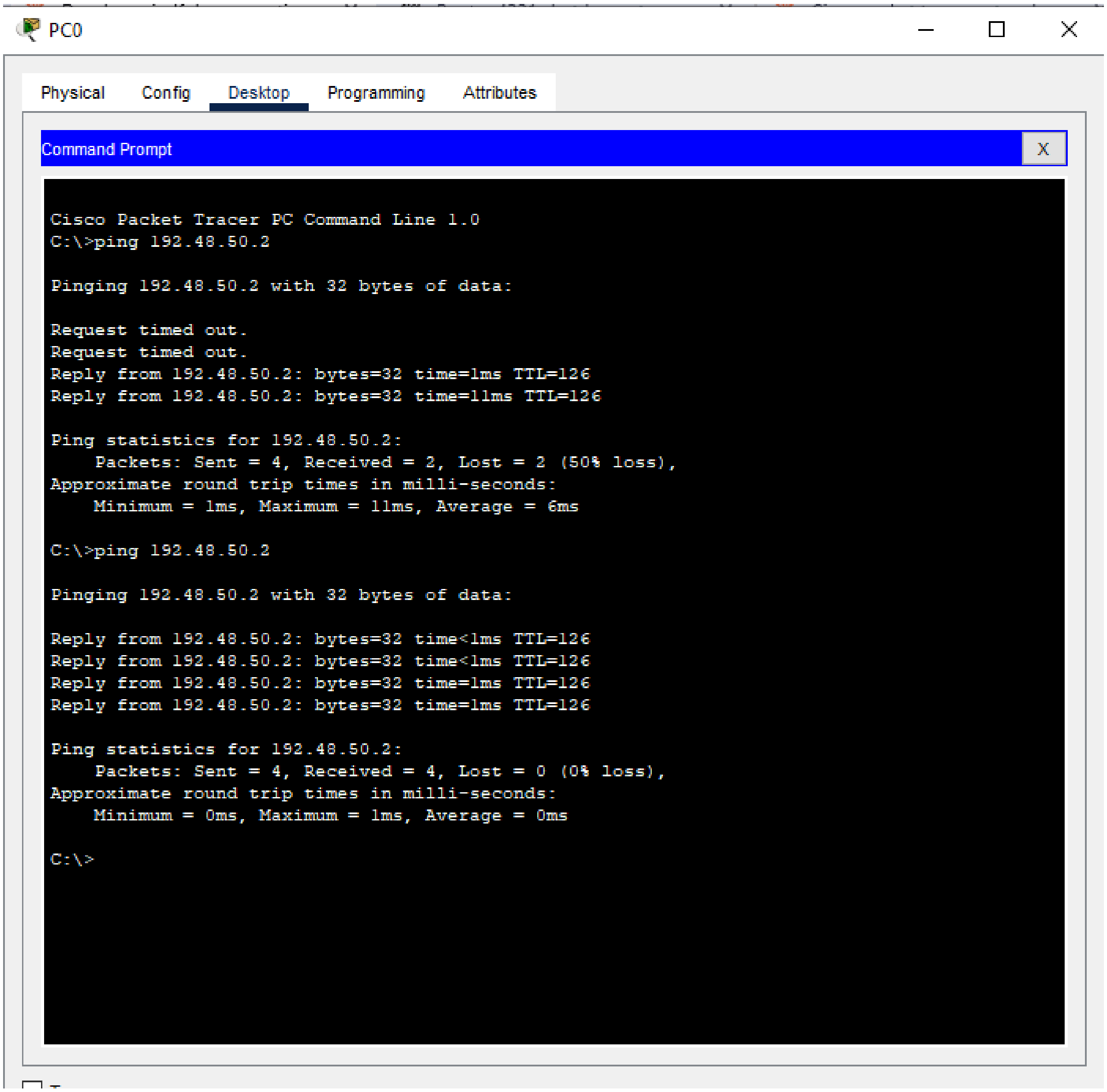


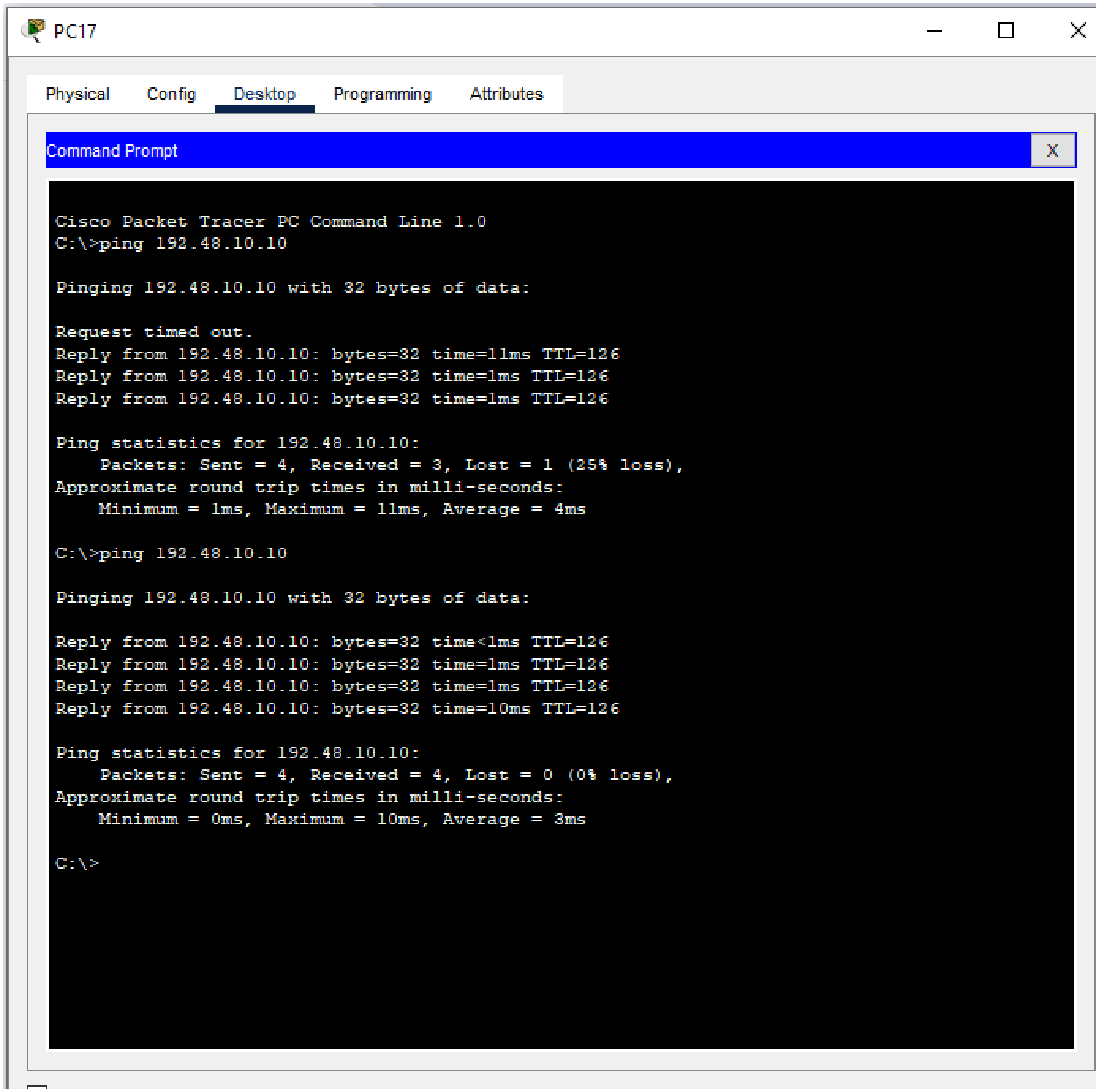
```
Fa0/13                       notconnect   20         auto    auto  10/100BaseTX
Fa0/14                       notconnect   20         auto    auto  10/100BaseTX
Fa0/15                       notconnect   20         auto    auto  10/100BaseTX
Fa0/16                       connected    30         auto    auto  10/100BaseTX
Fa0/17                       connected    30         auto    auto  10/100BaseTX
Fa0/18                       notconnect   30         auto    auto  10/100BaseTX
Fa0/19                       notconnect   30         auto    auto  10/100BaseTX
Fa0/20                       notconnect   30         auto    auto  10/100BaseTX
Fa0/21                       connected    40         auto    auto  10/100BaseTX

Switch5#
```

Block 4:

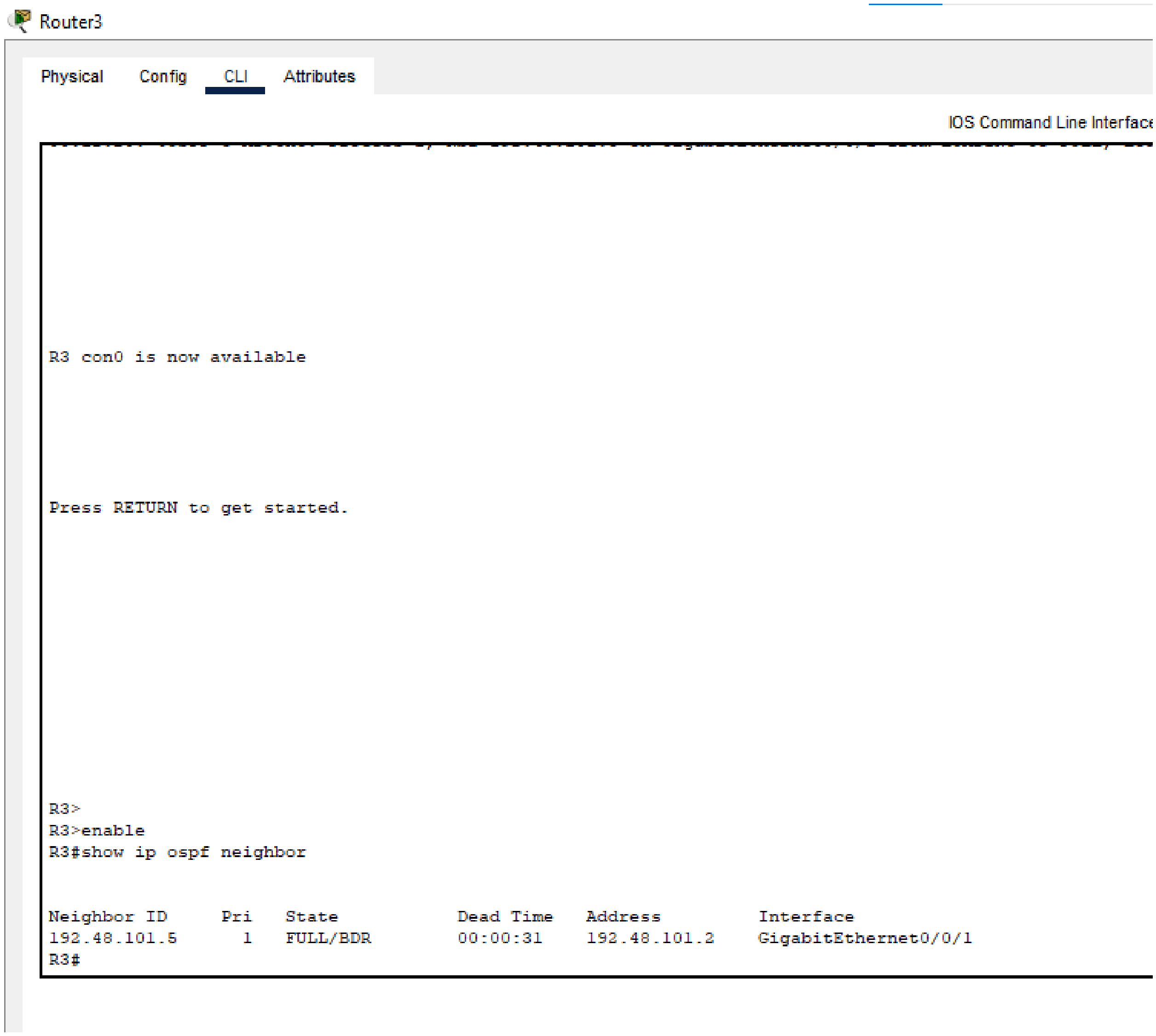

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
R1(config)#interface gigabitEthernet0/0/0
R1(config-if)#ip address 192.48.50.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
R1(config-if)#exit
R1(config)#access-list 10 permit host 192.48.50.2
R1(config)#access-limit 10 deny host 192.48.50.3
^
% Invalid input detected at '^' marker.
R1(config)#access-ist 10 deny host 192.48.50.3
^
% Invalid input detected at '^' marker.
R1(config)#access-list 10 deny host 192.48.50.3
R1(config)#access-list 10 permit any
R1(config)#interface gigabit
% Incomplete command.
R1(config)#interface gigabitEthernet 0/0/0
R1(config-if)#ip access-group 10 in
R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#show access-lists
Standard IP access list 10
10 permit host 192.48.50.2
20 deny host 192.48.50.3
30 permit any
R1#
Copy
Paste

Block 5:


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.48.50.2
Pinging 192.48.50.2 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.48.50.2: bytes=32 time=1ms TTL=126
Reply from 192.48.50.2: bytes=32 time=11ms TTL=126
Ping statistics for 192.48.50.2:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 11ms, Average = 6ms
C:\>ping 192.48.50.2
Pinging 192.48.50.2 with 32 bytes of data:
Reply from 192.48.50.2: bytes=32 time<1ms TTL=126
Reply from 192.48.50.2: bytes=32 time<1ms TTL=126
Reply from 192.48.50.2: bytes=32 time=1ms TTL=126
Reply from 192.48.50.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.48.50.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


PC17
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.48.10.10
Pinging 192.48.10.10 with 32 bytes of data:
Request timed out.
Reply from 192.48.10.10: bytes=32 time=11ms TTL=126
Reply from 192.48.10.10: bytes=32 time=1ms TTL=126
Reply from 192.48.10.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.48.10.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 11ms, Average = 4ms
C:\>ping 192.48.10.10
Pinging 192.48.10.10 with 32 bytes of data:
Reply from 192.48.10.10: bytes=32 time<1ms TTL=126
Reply from 192.48.10.10: bytes=32 time=1ms TTL=126
Reply from 192.48.10.10: bytes=32 time=1ms TTL=126
Reply from 192.48.10.10: bytes=32 time=10ms TTL=126
Ping statistics for 192.48.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms
C:\>

Block 6:


Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
R3 con0 is now available
Press RETURN to get started.
R3>
R3>enable
R3#show ip ospf neighbor
Neighbor ID   Pri   State   Dead Time   Address   Interface
192.48.101.5   1   FULL/BDR   00:00:31   192.48.101.2   GigabitEthernet0/0/1
R3#

Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
R4 con0 is now available




Press RETURN to get started.









R4>enable
R4#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.48.101.1      1   FULL/DR         00:00:31    192.48.101.1    GigabitEthernet0/0/0
192.48.101.9      1   FULL/BDR        00:00:35    192.48.101.6    GigabitEthernet0/0/1
R4#
```

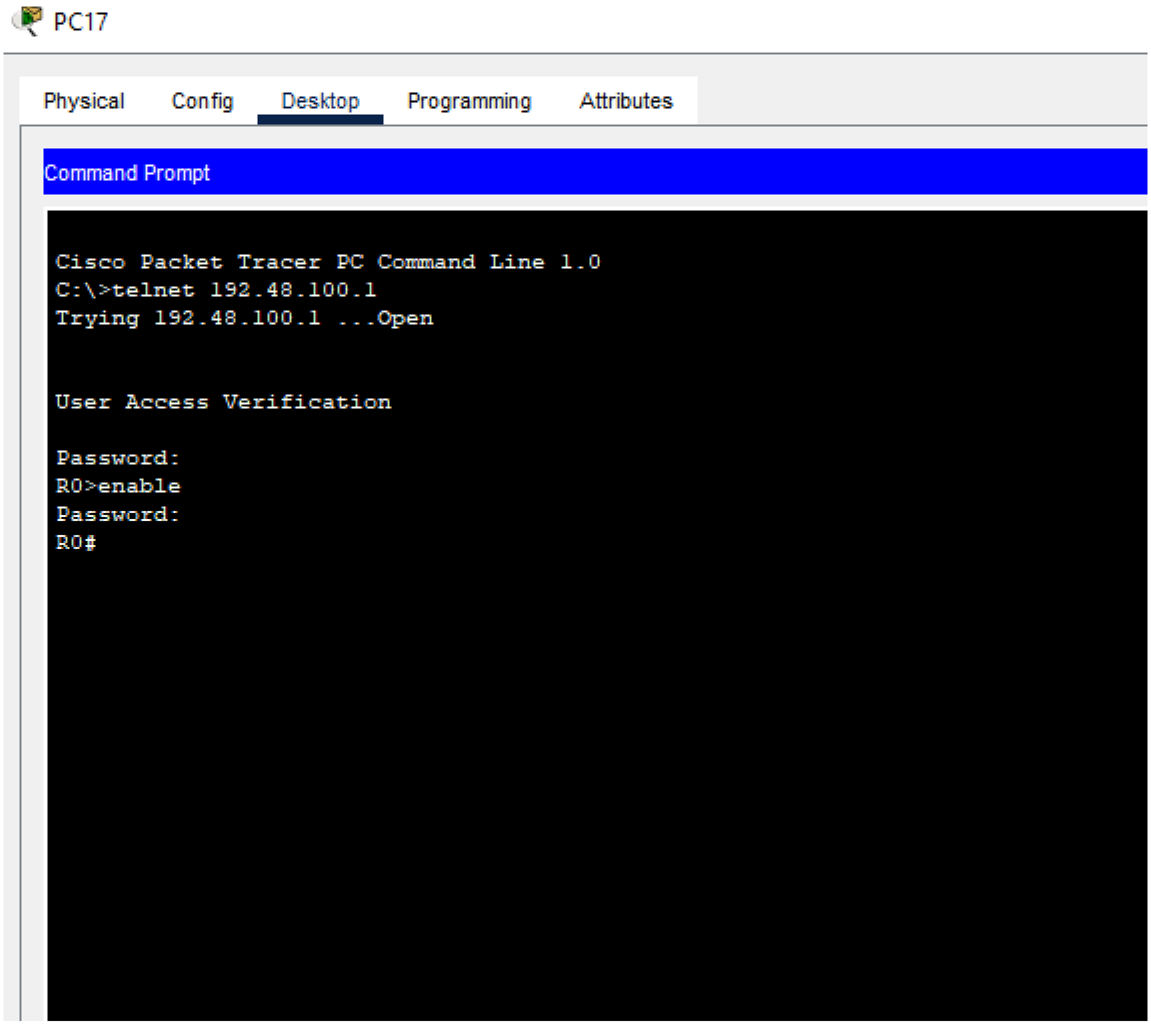
Copy Paste

```
Press RETURN to get started.









R5>enable
R5#show ip route ospf
     192.48.101.0/24 is variably subnetted, 5 subnets, 2 masks
O       192.48.101.0 [110/2] via 192.48.101.5, 00:11:47, GigabitEthernet0/0/0

R5#
```

```
R6#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.48.101.10
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.48.101.8 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.48.101.1         110      00:19:22
    192.48.101.5         110      00:12:32
    192.48.101.9         110      00:09:32
    192.48.101.10        110      00:09:32
  Distance: (default is 110)

R6#ping 192.48.101.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.48.101.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/16 ms

R6#
```

Copy

Block 8:


PC17
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.48.100.1
Trying 192.48.100.1 ...Open
User Access Verification
Password:
R0>enable
Password:
R0#